Assignment – M1

Objective:

Design the architecture and implement a QR code-based Attendance and Access Control System for a manufacturing company, adhering to SOLID principles, satisfying both functional and non-functional requirements, and providing software architecture documentation from multiple architectural views.

The project consists of two submission milestones (M1: focusing on requirements analysis and system design; M2: finalizing the system design, implementing and testing the system). Similar to a real-world software project, the project scope and requirements will be released incrementally through separate documents; details of subsequent phases will be announced when their respective milestones are reached.

Context:

TMC Manufacturing Joint Stock Company operates in the manufacturing sector and currently owns two factories: Binh Duong (~800 employees) and Long An (~600 employees).

The company intends to develop a system to control employee access to the workplace while also supporting the generation of monthly attendance reports.

The system uses QR codes for employee identification and attendance check-in/check-out. When an employee passes through an access gate, a QR scanner triggers a signal to the turnstile controller to unlock the gate. Each access event (entry or exit) is recorded as an attendance transaction.

Integration: The company already has an existing HR/Payroll System. Attendance data must be transmitted to this system through APIs and be available for reconciliation before 12:00 PM on the first day of each month. The company also has an existing turnstile control system, which exposes APIs for controlling gate opening and closing for each access event.

Users (Actors): employees, production supervisors, HR staff, security guards, system administrators, and executive management (for reporting and monitoring purposes). Every employee has a personal smartphone that can access the company’s mobile application, web portal, or email system and display a QR code.

Out of scope for this phase: visitor management; payroll calculation (the system only exports attendance data and does not perform payroll processing).

Scope of the System:

The key functional and non-functional requirements are listed below. Students may propose additional requirements if deemed appropriate.

1. Functional Requirements

- User and QR Code Management
- Gate Station Management
- Attendance Event Recording
- Reporting and Auditing
- System Administration

2. Non-Functional Requirements

- Kiosk response time ≤ 2 seconds under the current workload conditions (peak shift changes: approximately 800 access transactions within 30 minutes per factory).
- No loss of attendance records during normal system operation.
- Role-Based Access Control (RBAC) with comprehensive audit logging.
- Protection of employees' personal data in compliance with applicable data privacy regulations.

Assignment Tasks:

***Task 1*: Software Architecture Design**

1. Thoroughly describe the context of the system based on the basic information provided. This includes clearly identifying functional and non-functional requirements, outlining the objectives and scope of the project, etc.
2. Create the software architecture for the TMC: The software architecture should include:
    - Architecture Characteristics define the success criteria of the TMC.
    - Structure:
        - Compare and choose suitable architecture styles to apply to the TMC.
        - Present step by step how to create the logical and physical software architecture of the system.
    - Architecture Decisions define the rules for how the TMC should be constructed.

- Design Principles: Guidelines for constructing the TMC.

Submission Instructions:

- Submit a detailed report on the LMS website.